

Feedback-Dressed Dynamic Topology of Inverter-Dominated Power Systems: Hidden Cycles, Reinforcement Reversals, and Controller-Aware Screening

Jorge L. Mayorga Taborda
Universidad de los Andes, Bogotá, Colombia

Abstract—Conventional grid-strength screens describe the physical electrical network, whereas inverter-rich stability is determined by a closed-loop network that includes controller dynamics. This paper develops a controller-aware interpretation of that discrepancy. A controller self-energy $\Sigma(s)$ is composed with the frequency-dependent network resolvent $G_r(s)$ to form a *feedback-dressed network operator* $K_H(s) = G_r(s)\Sigma(s)$. The corresponding controller-side return ratio $R_c(s)$ has the same nonzero loop spectrum, and the full characteristic determinant factors exactly into local network dynamics, local controller dynamics, and an interaction determinant. Schur condensation then exposes frequency-dependent effective edges and cycles that need not exist in the retained physical graph. The framework is evaluated on a transparent IEEE 39-bus proof-of-concept: a MATPOWER AC operating point is coupled to a stylized grid-following converter at bus 34. Reinforcing branch 19–20 by $1.8\times$ improves algebraic connectivity by 0.694%, reduces a bus-34 network-impedance proxy by 4.60%, and reduces AC losses by 0.227%, yet collapses the network–controller feedback margin by 91.3% and reduces the controller-family damping ratio from 2.89% to 0.186%. A passive reduced cycle gain of 0.00596 becomes 1.013 after controller dressing and falls to 0.0787 after controller retuning while the physical reinforcement is retained. An exhaustive 46-branch action screen, frozen before nonlinear simulation, ranks three deceptive reinforcements whose nonlinear controller-stress amplification is ordered exactly as predicted. Fiedler support and the hidden dynamic red zone show low spatial overlap. The results suggest that controller-aware planning should screen not only physical strength but also the dynamic feedback topology created by network propagation and controller self-energy.

Index Terms—inverter-based resources, grid-following inverter, small-signal stability, graph signal processing, Fiedler vector, Schur complement, self-energy, hidden dynamic graph, feedback cycle, transmission planning.

I. INTRODUCTION

LOW-INERTIA and inverter-dominated grids are often screened with quantities derived from the physical network: short-circuit strength, impedance, loading, algebraic connectivity, Laplacian eigenvectors, or reduced L – M – D models. These quantities are useful because they summarize the network seen by power-flow or electromechanical coordinates. They are not, however, a complete representation of the

feedback paths created by phase-locked loops (PLLs), current controllers, outer loops, exciters, governors, stabilizers, and other dynamic devices.

Impedance-based stability criteria already establish that grid and converter dynamics must be analyzed jointly [2], and interaction-admittance models extend that view to multi-inverter networks [3]. PLL–grid coupling is also known to change small-signal stability and can require controller-side remedies [4]. Separately, Kron reduction formalizes how eliminating electrical nodes creates effective edges in reduced networks [5], and graph signal processing (GSP) provides a spectral language for signals and operators on graph modes [7]. The question pursued here is narrower: *how should a physical topology action be mapped to the closed-loop dynamic feedback that actually determines an inverter-rich stability margin?*

The key observation is that controller dynamics are not stability-relevant merely because they exist. Their effect must be propagated through the dynamic susceptibility of the network and returned through the feedback loop. We therefore distinguish four objects:

$$L_P \longrightarrow \Sigma(s) \longrightarrow K_H(s) = G_r(s)\Sigma(s) \longrightarrow R_c(s), \quad (1)$$

where L_P describes the physical graph, $\Sigma(s)$ is the controller self-energy on retained network states, $G_r(s)$ is the network resolvent, $K_H(s)$ is the feedback-dressed network operator, and $R_c(s)$ is the controller-side return ratio. The proposed *hidden margin* is based on the distance of $I - R_c(j\omega)$ from singularity.

The algebra used below—Schur complements, determinant factorizations, Woodbury updates, and spectral graph projections—is standard. The proposed contribution is their specialization into a single power-system workflow from *physical action* to *hidden feedback topology* to *dynamic consequence*. The main contributions are:

- an exact determinant factorization that separates local network and controller dynamics from interaction-induced modes;
- a feedback-dressed effective graph whose hidden edges and cycles are generated by Schur condensation and controller/network propagation;

- a low-rank topology-to-feedback map for transmission actions, together with a controller-side hidden feedback margin;
- a GSP interpretation showing why a Fiedler-only or raw- Σ view is insufficient, while feedback-dressed cross-mode coupling is more informative;
- an exhaustive IEEE 39-bus proof-of-concept in which electrically beneficial reinforcements are ranked before nonlinear simulation and the frozen ranking predicts nonlinear controller stress.

The paper intentionally does not claim independent DAE/EMT validation. MATPOWER [8] is used for the AC operating point and physical-network metrics; ANDES [9] is identified as the next independent validation environment.

II. PHYSICAL AND DYNAMIC NETWORK MODELS

A. Physical AC graph and Fiedler coordinates

Let a solved AC operating point be denoted by $x^* = (\theta^*, V^*, \dots)$. A local physical coupling matrix can be formed from the angle Jacobian or, in the lossless approximation,

$$L_P(x^*) = B W(x^*) B^\top, \quad W_{ee} = b_e V_i^* V_j^* \cos(\theta_i^* - \theta_j^*), \quad (2)$$

where B is the branch incidence matrix. Its ordered eigenpairs satisfy

$$L_P U = U \Lambda, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots, \quad (3)$$

with λ_2 the algebraic connectivity and u_2 the Fiedler vector [6]. The graph Fourier transform of a bus signal y is $\hat{y} = U^H y$ [7]. These coordinates provide a natural baseline for asking whether controller dynamics preserve or mix the physical graph modes.

B. Linearized DAE and controller self-energy

Consider a linearized dynamic model after algebraic elimination,

$$\dot{x} = A x, \quad A = \begin{bmatrix} A_{rr} & A_{rc} \\ A_{cr} & A_{cc} \end{bmatrix}, \quad (4)$$

where r denotes retained network/interface states and c controller/device states. Define

$$T_r(s) = sI - A_{rr}, \quad T_c(s) = sI - A_{cc}, \quad (5)$$

and the network-side controller self-energy

$$\Sigma_r(s) = A_{rc} T_c(s)^{-1} A_{cr}. \quad (6)$$

The exact retained operator is

$$T_{\text{eff},r}(s) = T_r(s) - \Sigma_r(s). \quad (7)$$

For electromechanical retained coordinates, (7) contains the familiar second-order form $s^2 M + s \Sigma_D(s) + L$ as a special representation, but the state-space form is used here because it does not require controller dynamics to be collapsed into a scalar damping coefficient.

C. Feedback-dressed network operator and hidden margin

Let

$$G_r(s) = T_r(s)^{-1}, \quad G_c(s) = T_c(s)^{-1}. \quad (8)$$

We define the feedback-dressed network operator

$$K_H(s) = G_r(s) \Sigma_r(s) = G_r A_{rc} G_c A_{cr} \quad (9)$$

and the controller-side return ratio

$$R_c(s) = G_c(s) A_{cr} G_r(s) A_{rc}. \quad (10)$$

The proposed hidden feedback margin is

$$d_H = \min_{\omega \in \Omega} \sigma_{\min}[I - R_c(j\omega)]. \quad (11)$$

Unlike a physical strength index, d_H is a closed-loop quantity: it becomes small when a network-mediated controller return approaches singular feedback closure.

III. ALGEBRAIC STRUCTURE OF HIDDEN DYNAMIC TOPOLOGY

A. General interaction determinant

Proposition 1 (Interaction determinant): For frequencies where $T_r(s)$ and $T_c(s)$ are nonsingular,

$$\det(sI - A) = \det T_r(s) \det T_c(s) \det[I - R_c(s)]. \quad (12)$$

Equivalently,

$$\det(sI - A) = \det T_c(s) \det T_r(s) \det[I - K_H(s)]. \quad (13)$$

Proof: Apply the Schur determinant identity to $sI - A$ using either diagonal block. The second equality follows from Sylvester's determinant theorem $\det(I - XY) = \det(I - YX)$. ■

Equation (12) separates local poles of the network and controller blocks from interaction-induced zeros of $\det(I - R_c)$. This is the precise sense in which the proposed hidden topology describes dynamics that are not visible from the physical Laplacian alone.

Proposition 2 (Network/controller spectral equivalence): $K_H(s)$ and $R_c(s)$ have identical nonzero eigenvalues, including algebraic multiplicities.

Proof: Write $X = G_r A_{rc}$ and $Y = G_c A_{cr}$. Then $K_H = XY$ and $R_c = YX$. The nonzero spectra of XY and YX coincide. ■

This result justifies analyzing the interaction in either network coordinates (useful for GSP and localization) or controller coordinates (useful for low-dimensional loop screening).

B. Hidden effective edges and cycles

Partition a dynamic operator into retained ports b and hidden states h :

$$T(s) = \begin{bmatrix} T_{bb} & T_{bh} \\ T_{hb} & T_{hh} \end{bmatrix}. \quad (14)$$

Eliminating h gives

$$T_{\text{eff}}(s) = T_{bb} - T_{bh} T_{hh}^{-1} T_{hb}. \quad (15)$$

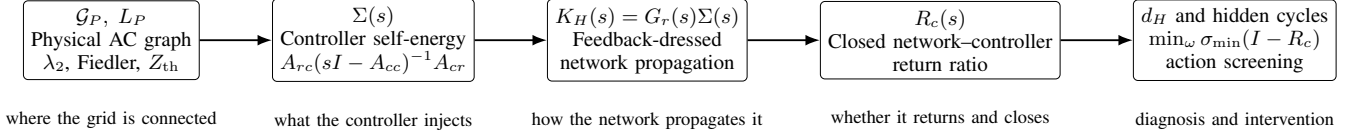


Fig. 1. Hierarchy used in this paper. The physical Laplacian is a structural description; controller self-energy becomes stability-relevant after propagation through the network resolvent and closure of the network-controller loop.

An off-diagonal element of T_{eff} can be nonzero even when the corresponding retained ports have no direct physical edge. Such an element is called an *effective edge*; if it changes only after controller dressing it is controller-mediated rather than a purely passive Kron edge.

For a three-port normalized operator $T_{\text{eff}} = D(I - W)$ with $\text{diag}(W) = 0$, direct determinant expansion gives

$$\det(I - W) = 1 - W_{12}W_{21} - W_{13}W_{31} - W_{23}W_{32} - W_{12}W_{23}W_{31} - W_{13}W_{32}W_{21}. \quad (16)$$

Thus the interaction determinant is exactly decomposed into three directed 2-cycles and two oriented 3-cycles. For larger scalar-port graphs, the Leibniz expansion yields signed collections of vertex-disjoint cycles, while the log-determinant trace expansion organizes closed walks.

C. Rank-one cycle closure and topology updates

When $A_{rc} = bq^T$ and $A_{cr} = uc^T$, the return ratio has rank at most one. Its only possible nonzero eigenvalue is

$$h(s) = \underbrace{q^T G_c(s) u}_{G_c^{\text{loop}}(s)} \underbrace{c^T G_r(s) b}_{G_n^{\text{loop}}(s)}, \quad (17)$$

and

$$\det(sI - A) = \det T_r \det T_c [1 - h(s)]. \quad (18)$$

Hence an interaction-induced mode satisfies $h(s) = 1$ when s is not already a local pole.

At a frozen operating point, let a physical action update the network operator by rank one,

$$T_r(p) = T_r + p u_e v_e^T. \quad (19)$$

Sherman–Morrison gives

$$G_r(p) = G_r - \frac{p G_r u_e v_e^T G_r}{1 + p v_e^T G_r u_e}. \quad (20)$$

Substitution into (17) yields the exact finite-action map

$$h(p, s) = h(0, s) - \frac{p G_c^{\text{loop}}(s) [c^T G_r u_e] [v_e^T G_r b]}{1 + p v_e^T G_r u_e}. \quad (21)$$

For a full AC action, (21) is the direct frozen-point contribution; the total derivative also contains the operating-point shift from the implicit-function theorem.

D. GSP view of controller-mediated mode mixing

Let U be the physical Laplacian basis. A raw controller-dressing map is

$$\hat{\Sigma}(j\omega) = U^H \Sigma_{\text{bus}}(j\omega) U. \quad (22)$$

Its off-diagonal terms quantify controller-induced mixing between physical graph modes. The experiments below show that raw $\hat{\Sigma}$ is not sufficient. The more informative object is the feedback-dressed projection

$$\hat{K}_H(j\omega) = U^H K_{H,\text{bus}}(j\omega) U, \quad (23)$$

which includes both controller dressing and frequency-dependent network propagation. We use

$$g_{\max} = \max_{k \neq \ell} |[\hat{K}_H]_{k\ell}| \quad (24)$$

as an interpretable cross-mode diagnostic rather than as a replacement for the exact hidden margin (11).

IV. IEEE 39-BUS PROOF-OF-CONCEPT

A. Model and numerical protocol

The physical operating point is the standard MATPOWER IEEE 39-bus case [8]. A Newton–Raphson AC power flow implemented directly from the case data reproduces the stored MATPOWER solution to a maximum voltage-magnitude error of 4.8×10^{-8} p.u., a maximum angle error of 5.0×10^{-7} degrees, and a power-flow residual of 1.4×10^{-13} p.u.

To expose the proposed mechanism transparently, the dynamic proof-of-concept uses a stylized grid-following converter (GFL) at bus 34 with PLL angle/frequency states, a power/current-control state, and a hard current limit in the nonlinear stress test. The remaining network uses reduced swing/frequency-sensitive dynamics. This surrogate is intentionally simpler than a production-grade renewable model; its role is to test the algebraic mechanism, exhaustive action ranking, and falsifiable predictions before independent DAE/EMT validation.

Each of the 46 existing branches is reinforced independently by multiplying its susceptance by 1.8. The AC operating point is re-solved for every action. Physical metrics include algebraic connectivity, an effective bus-34 network-impedance proxy, and AC active losses. Dynamic metrics include the critical pole, d_H , hidden effective-cycle gains, Fiedler/hidden-mode spatial support, and GSP projections. A dangerous/benign ranking is frozen before a broadband nonlinear Ornstein–Uhlenbeck stress is applied. A separate resonance-matched 500-trajectory test evaluates access to the current-limit boundary in the star case.

TABLE I
STAR-CASE METRICS FOR BRANCH 19–20

Metric	Base	1.8× action
$\Delta\lambda_2$	–	+0.694%
Bus-34 impedance proxy	–	–4.60%
AC active losses	–	–0.227%
Feedback-margin change	–	–91.3%
Controller-family damping	2.89%	0.186%
Dominant loop eigenvalue	–	0.998027 – j0.000568

B. Star action: reinforcement reversal on branch 19–20

Table I summarizes the central counterexample. At 1.8×, the physical network improves by three conventional measures, while the dynamic feedback margin collapses by more than 90% and the controller-family damping drops to 0.186%. At 2.0×, approximately corresponding to an identical parallel-circuit equivalent in this surrogate, the critical controller-family pole crosses the imaginary axis.

C. Hidden cycle and causal controller ablation

Retaining only buses {16, 33, 34} creates a stringent hidden-topology test because the retained triplet has no direct physical lines among its members. Schur condensation of the remaining states produces a dense frequency-dependent effective graph. At 1.417 Hz, the effective 2-cycle $33 \leftrightarrow 34$ has gain 0.00596 in the reinforced passive network, 1.013 in the reinforced controller-dressed model, and 0.0787 after controller retuning while the physical reinforcement is unchanged. The result distinguishes a controller-mediated effective cycle from a passive Kron artifact.

D. Exhaustive action atlas and frozen nonlinear cross-check

All 46 existing branches are screened at 1.8×. Among electrically beneficial actions, the physical connectivity change alone does not determine the closed-loop feedback margin. The frozen three most severe feedback-margin collapses are 19–20, 19–33, and 16–19. Before nonlinear simulation, three benign controls (8–9, 29–38, and 4–5) are also frozen. All six cases then receive the same broadband stochastic stress and seed.

The predicted dangerous actions increase median controller-action RMS by 52.9%, 32.5%, and 10.1%, respectively; the three predicted benign actions remain essentially unchanged. Across the six frozen actions, the rank correlation between predicted feedback-margin collapse and nonlinear RMS amplification is $\rho = 1.00$. The nonlinear event-probability ordering gives $\rho = 0.941$ ($p = 0.0051$). A half-step integration check preserves the dangerous/benign ordering.

A separate resonance-matched nonlinear test uses 500 trajectories per case over 20 s. Sustained current-limit entry is observed in 19.0% of base trajectories, 61.0% after the 1.8× 19–20 reinforcement, and 0/500 after retuning. These absolute probabilities depend on the surrogate, forcing spectrum, limiter, and horizon; only the relative mechanism is interpreted.

TABLE II
FROZEN ACTION RANKING AND NONLINEAR CROSS-CHECK

Action	Δd_H (%)	RMS amp. (%)	Event prob.
19–20	–91.3	+52.9	0.115
19–33	–62.0	+32.5	0.050
16–19	–23.9	+10.1	0.020
8–9	≈ 0	≈ 0	0.010
29–38	≈ 0	≈ 0	0.010
4–5	≈ 0	≈ 0	0.010

E. Fiedler weak zones versus hidden dynamic red zones

The physical Fiedler support and dynamic support differ substantially. For the 19–20 action, the cosine overlap between Fiedler support and the critical hidden-mode support is 0.280; for 16–19 it is 0.145. In the star case, the critical hidden mode is concentrated on bus 34 (88.7%), bus 20 (9.0%), and bus 19 (2.0%), whereas the Fiedler support is distributed across a broader physical cut structure. This result should not be read as a failure of Fiedler analysis: it answers a different question. Fiedler coordinates describe physical connectivity; the hidden dynamic red zone localizes the closed-loop mechanism that sets the dynamic margin.

F. What the GSP experiment changed

A useful negative result emerged from the GSP study. Raw mixing of physical graph modes by $\hat{\Sigma} = U^H \Sigma U$ is only weakly associated with the hidden margin ($\rho = 0.275$, $p = 0.064$). A Fiedler-only dynamic projection is essentially uninformative ($\rho = 0.058$). Once the controller self-energy is propagated through the network resolvent, however, the strongest cross-graph-mode coupling of $\hat{K}_H$ is significantly associated with hidden fragility ($\rho = -0.415$, $p = 0.0041$). The full-loop spectral radius is the strongest compact representation in this model ($\rho = -0.865$, $p = 9.4 \times 10^{-15}$), consistent with the exact determinant factorization.

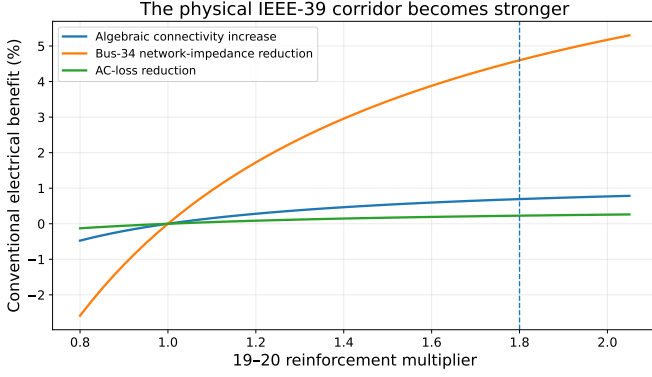
V. PLANNING AND INDUSTRIAL INTERPRETATION

The framework produces a decision structure that differs from a conventional weak-bus ranking. A candidate action can improve the physical network and still reduce the hidden margin. We refer to this situation as a *controller-mediated reinforcement reversal*. The operator should not necessarily reject the physical action: the causal ablation suggests a more useful workflow,

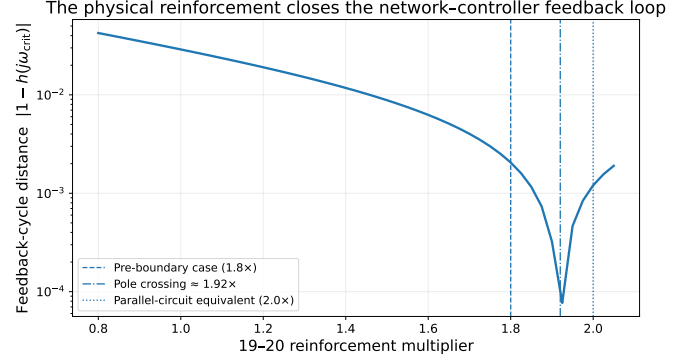
$$\text{keep reinforcement} + \text{retune limiting controller.} \quad (25)$$

This interpretation supports several industrial uses.

- 1) **Transmission screening:** use AC metrics to remove physically unattractive projects, then use d_H and hidden-cycle diagnostics to identify actions requiring detailed dynamic studies.
- 2) **Commissioning:** determine whether a topology change requires PLL/current-loop retuning before energization.
- 3) **IBR/BESS siting:** compare candidate locations by their effect on both physical strength and hidden-cycle closure rather than only SCR or Fiedler participation.

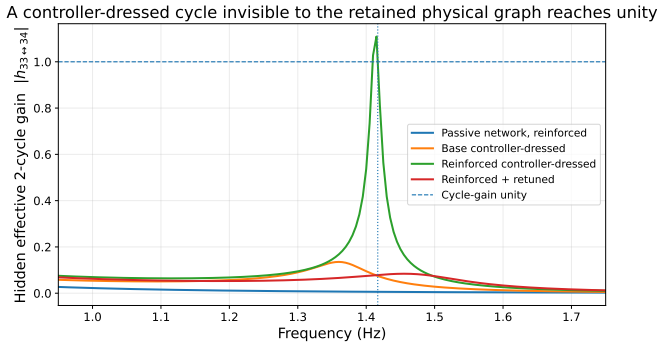


(a) Physical-network metrics improve monotonically.

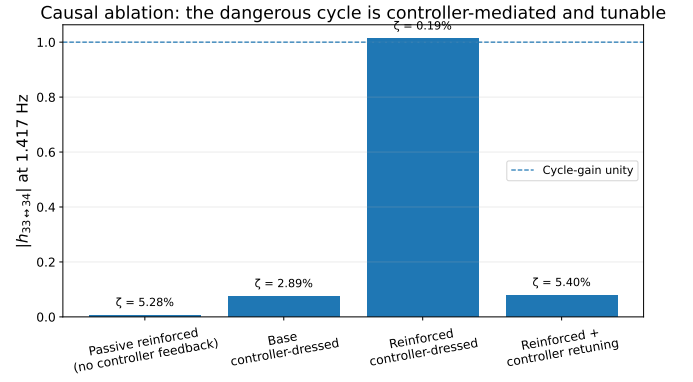


(b) The network-controller loop approaches closure.

Fig. 2. Reinforcement reversal on branch 19–20. The physical grid becomes stronger while the closed-loop dynamic margin deteriorates.



(a) Frequency dependence of the hidden effective cycle.



(b) Causal ablation: passive, dressed, and retuned cases.

Fig. 3. The critical effective cycle is small in the passive reduction, becomes near-unity after controller dressing, and is suppressed by retuning without undoing the grid reinforcement.

- 4) **Dynamic red zones:** project critical hidden modes and effective cycles back onto buses, lines, and devices to identify the physical support of the margin-setting feedback sector.
- 5) **Model reduction:** reduce large regions to port operators while preserving the feedback-dressed cycles that control stability rather than preserving only static terminal behavior.

In industrial facilities, the same logic can be applied to a PCC feeding active front ends, variable-frequency drives, UPS systems, filters, PV inverters, and BESS. The physical one-line diagram may be radial, yet controller/network condensation can create an effective loop among devices. The relevant engineering question becomes not only “how strong is the PCC?” but “which controller-mediated loop is closest to closure at the frequencies excited by the installed equipment?”

VI. RELATION TO PRIOR WORK

The proposed framework is complementary to impedance-based and interaction-admittance stability analysis, not a replacement. Sun’s impedance criterion [2] and subsequent multivariable approaches establish the importance of converter/grid return ratios. Lu *et al.* [3] provide a physical admittance interpretation of multi-inverter interaction, while

PLL/grid coupling and controller-side remedies are demonstrated in [4]. The present work differs by making the *planning action* explicit and by tracking how that action changes a frequency-dependent effective topology, its hidden cycles, and its physical support.

Similarly, Kron reduction [5] explains why passive node elimination produces effective edges. The controller-dressed effective edges used here include that passive contribution but are separated experimentally from it. GSP [7] supplies the modal coordinates used to compare Fiedler structure with dynamic feedback mixing; the key empirical result is that raw graph modes or raw controller self-energy are not sufficient without the network resolvent. Finally, nonlinear synchronization results in oscillator networks [10] motivate graph-based reasoning for power-system dynamics, but the present object is a frequency-dependent closed-loop operator rather than a fixed physical coupling graph.

VII. LIMITATIONS AND FALSIFIABLE NEXT TESTS

The results are intentionally framed as a proof-of-concept. The dynamic model is not a validated production GFL model; the reported stochastic event probabilities are not outage probabilities; and the controller parameters were chosen to

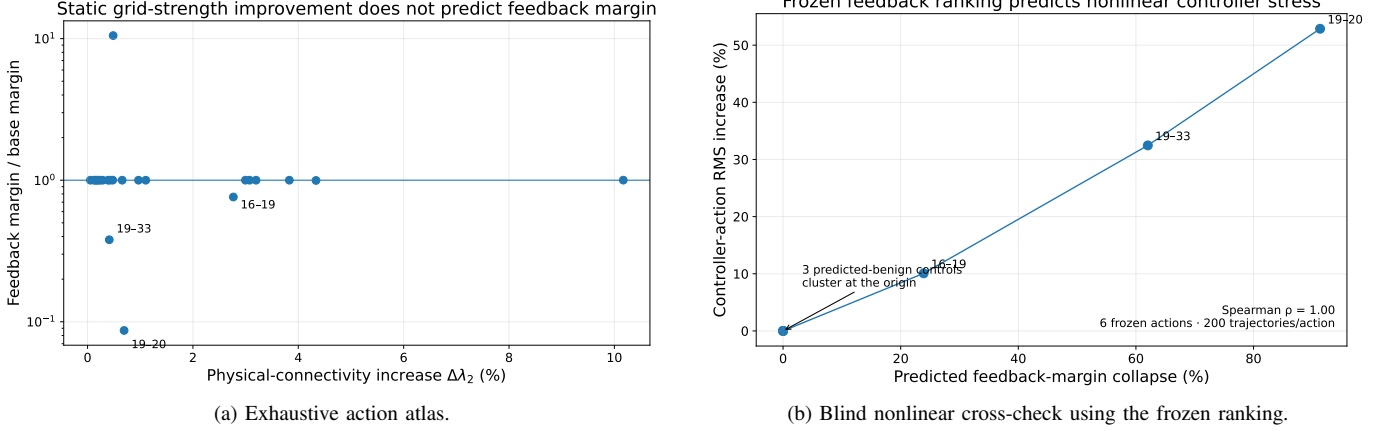


Fig. 4. The controller-aware screen is used before nonlinear simulation. This avoids selecting the star action after observing nonlinear behavior.

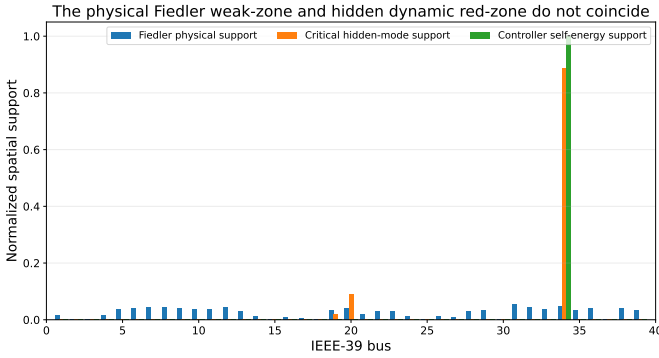


Fig. 5. Physical Fiedler support, critical hidden-mode support, and controller self-energy support for the 19–20 star action.

expose the mechanism transparently. The decisive next test is independent validation with detailed DAE models.

The ranking has therefore been frozen before such validation: 19–20, 19–33, and 16–19 are the predicted dangerous actions. In ANDES, the same ranking should be tested without re-selection using REGCP1/PLL dynamics, full power-flow/DAE relinearization, and time-domain disturbances. The falsifiable outcomes are: (i) whether the frozen ranking predicts damping degradation, (ii) whether d_H predicts critical-clearing-time or current/PLL excursion, and (iii) whether controller retuning restores transient margin while the physical reinforcement remains. EMT validation should follow only for the strongest cases.

A second extension is to evaluate $K_H(j\omega, x(t))$ along nonlinear trajectories. If hidden-cycle closure consistently precedes current limiting or loss of synchronization, the framework could become a transient early-warning variable. That extension is not claimed in the present paper.

VIII. CONCLUSION

This paper argues that inverter-rich grids should be viewed through two complementary topologies: the physical topology that carries power and a frequency-dependent feedback-dressed topology that carries dynamic return. The controller

self-energy $\Sigma(s)$ is necessary but not sufficient; the stability-relevant object is obtained after propagation through the network resolvent and closure of the network–controller loop. Exact determinant identities connect that object to interaction-induced modes, while Schur condensation exposes hidden effective edges and cycles.

In the IEEE 39-bus proof-of-concept, an electrically beneficial transmission reinforcement improves algebraic connectivity, network impedance, and AC losses while collapsing a controller-feedback margin and nearly eliminating damping. The critical effective cycle is negligible in the passive reduction, becomes near-unity after controller dressing, and is suppressed by controller retuning without removing the physical reinforcement. An exhaustive action screen frozen before nonlinear simulation predicts controller-stress ordering, whereas physical λ_2 , Fiedler projection, and raw self-energy mixing are substantially less informative. These results motivate a controller-aware planning workflow in which physical strength is screened first and hidden dynamic topology is then used to identify deceptive reinforcements, critical devices, and targeted control remedies.

APPENDIX A COORDINATE INVARIANCE OF HIDDEN-STATE ELIMINATION

Let the hidden coordinates be transformed by $x'_h = Sx_h$ with nonsingular S . The transformed blocks satisfy $T'_{bh} = T_{bh}S^{-1}$, $T'_{hb} = ST_{hb}$, and $T'_{hh} = ST_{hh}S^{-1}$. Therefore

$$T'_{bh}(T'_{hh})^{-1}T'_{hb} = T_{bh}T_{hh}^{-1}T_{hb}. \quad (26)$$

Hence the retained effective operator (15) is invariant to arbitrary invertible coordinate changes internal to the eliminated subsystem. Hidden effective edges defined at the retained ports are therefore not artifacts of the controller-state basis.

APPENDIX B SUBGRAPH COMPOSITION VIEW

Let $T_0(s) = \text{diag}(T_1(s), \dots, T_m(s))$ contain local subgraph operators and let $C(s)$ encode port couplings. Then

$$T = T_0 - C = T_0(I - G_0C), \quad G_0 = T_0^{-1}, \quad (27)$$

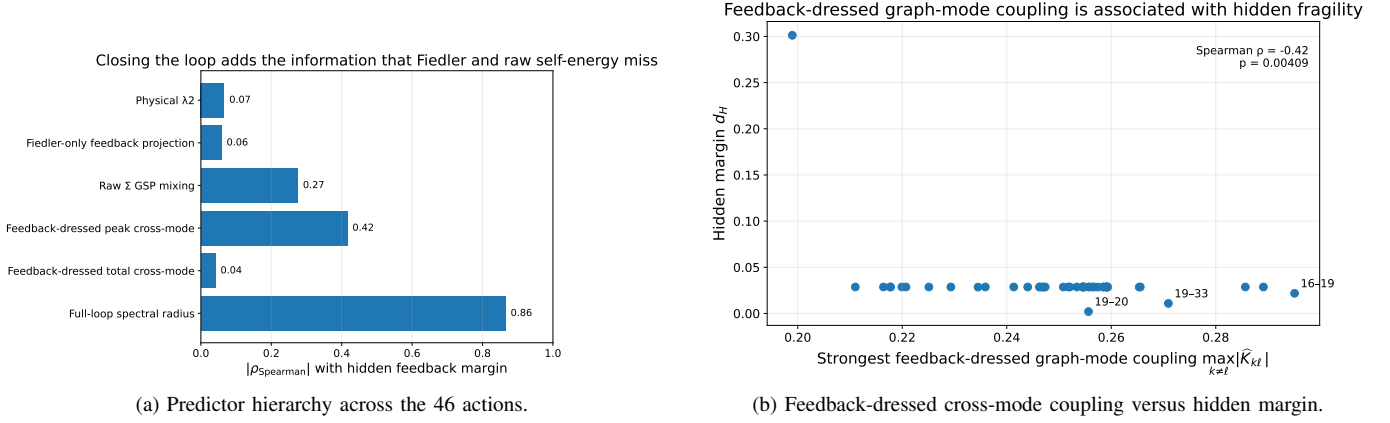


Fig. 6. The missing information is not another static graph metric. Stability-relevant modal mixing appears after controller self-energy is dressed by the network resolvent and closed through the feedback loop.

so that

$$T^{-1} = (I - G_0 C)^{-1} G_0. \quad (28)$$

If $\rho(G_0 C) < 1$,

$$T^{-1} = G_0 + G_0 C G_0 + G_0 C G_0 C G_0 + \dots \quad (29)$$

The terms are ordered subgraph walks. Closed products contribute to the interaction determinant $\det(I - G_0 C)$, providing a direct algebraic bridge between subgraph composition and feedback cycles.

REFERENCES

- [1] P. Kundur, *Power System Stability and Control*. New York, NY, USA: McGraw-Hill, 1994.
- [2] J. Sun, "Impedance-based stability criterion for grid-connected inverters," *IEEE Trans. Power Electron.*, vol. 26, no. 11, pp. 3075–3078, Nov. 2011, doi: 10.1109/TPEL.2011.2136439.
- [3] M. Lu, Y. Yang, B. Johnson, and F. Blaabjerg, "An interaction-admittance model for multi-inverter grid-connected systems," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 7542–7557, Aug. 2019, doi: 10.1109/TPEL.2018.2881139.
- [4] X. Lin, J. Yu, R. Yu, J. Zhang, Z. Yan, and H. Wen, "Improving small-signal stability of grid-connected inverter under weak grid by decoupling phase-lock loop and grid impedance," *IEEE Trans. Ind. Electron.*, vol. 69, no. 7, pp. 7040–7053, Jul. 2022, doi: 10.1109/TIE.2021.3095791.
- [5] F. Dörfler and F. Bullo, "Kron reduction of graphs with applications to electrical networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 60, no. 1, pp. 150–163, Jan. 2013, doi: 10.1109/TCSI.2012.2215780.
- [6] M. Fiedler, "Algebraic connectivity of graphs," *Czechoslovak Math. J.*, vol. 23, no. 2, pp. 298–305, 1973, doi: 10.21136/CMJ.1973.101168.
- [7] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandenheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, May 2013, doi: 10.1109/MSP.2012.2235192.
- [8] R. D. Zimmerman, C. E. Murillo-Sánchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 12–19, Feb. 2011, doi: 10.1109/TPWRS.2010.2051168.
- [9] H. Cui, F. Li, and K. Tomsovic, "Hybrid symbolic-numeric framework for power system modeling and analysis," *IEEE Trans. Power Syst.*, vol. 36, no. 2, pp. 1373–1384, Mar. 2021, doi: 10.1109/TPWRS.2020.3017019.
- [10] F. Dörfler, M. Chertkov, and F. Bullo, "Synchronization in complex oscillator networks and smart grids," *Proc. Natl. Acad. Sci. USA*, vol. 110, no. 6, pp. 2005–2010, Feb. 2013, doi: 10.1073/pnas.1212134110.